

MSFS File System

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Overview

This is a very simple file system with file and directory commands such as touch cat ,mkdir cd , ls and ect.

Structure

1. **Disk:** everything will be stored in disk image with .msfs format.Disk image contain all data and file system information.

Disk know as FileSystem Object in code and have this field :

```
final int TOTAL_BLOCKS;  
  
Block[] blocks;  
  
BitSet bitMap;  
  
Directory rootDirectory = new Directory("/", null);  
  
Directory currentDirectory = rootDirectory;
```

bitMap help us to know which block in blocks is free to write file data

Blocks data will be store in blocks

currentDirectory use for directory commands

```

public FileSystem(int size) {

    TOTAL_BLOCKS = size / 256;

    blocks = new Block[TOTAL_BLOCKS];

    bitMap = new BitSet(TOTAL_BLOCKS);

    for (int i = 0; i < TOTAL_BLOCKS; i++) {

        blocks[i] = new Block();

    }

}

```

The Constructor of Disk just gives size and creates blocks of size/ 256 because data will store in 256 bytes cells.

2. **Directory**: in Directory object we know the name of directory , path , map of files , parent Directory and map of sub Directories

```

class Directory implements Serializable {

    private String name;

    public String path = "/";

    private Map<String, File> files = new HashMap<>();

    private Map<String, Directory> subDirectories = new HashMap<>();

    private Directory parent;
}

```

3. **File:** in Directory

```
class File implements Serializable {

    private String fileName;

    private List<Integer> blockPointers = new ArrayList<>();

    private int size;
```

blockPointers tell us which blocks contains this file data

Execute MSFS app

MSFS programmed with java code. There is a bash script app in the directory to run MSFS apps .

First if we don't have any disk image . we must create one.

You must write the create command in the argument of MSFS app .

Create [disk_name] [size] [unit]

```
else if (args[0].equals("create") && v == 4) {

    fs = Helper.loadFileSystem(args[1]);

    if (fs != null) {

        System.out.println(args[1] + " disk is already exist and now open!");

    } else {

        fs = new FileSystem(Helper.byteSize(args[2], args[3]));

        Helper.saveFileSystem(args[1], fs);

        System.out.println(args[1] + " disk created!");

    }

}
```

After that we have a disk image on where MSFS runs .

Next time if we want work on exist disk we use mount instead of create:

mount [disk_name]

```
if (args[0].equals("mount") && v == 2) {  
  
    fs = Helper.loadFileSystem(args[1]);  
  
    if (fs == null) {  
  
        return;  
  
    }  
  
}
```

Now the program starts in the root directory of our disk image:

```
mounting disk image ....  
mmd2 disk is already exist and now open!  
MSFS>>mmd2.disk/
```

Directory commands

1. Mkdir

To create new directory in current directory

2. Cd

Change directory. If you want to go back to the parent directory use cd ..

3. Ls

Show list of directories and files in current directory . use ls -s to show the size of files

4. Rm

Remove directory

```
DirectoryHelper
```

This class is helper class and have some methods to these commands:

```
public static void removeDirectory(FileSystem fs, String dirName)

public static void makeDirectory(FileSystem fs, String dirName)

public static void listDirectorySize(FileSystem fs)

public static void listDirectory(FileSystem fs)

public static void changeDirectory(FileSystem fs, String dirName)
```

File commands

1. touch

To create new files or write to them.

2. Rmf

To remove files.

3. Cat

Show data of file.

```
class FileHandleHelper
```

This class is helper class and have some methods to these commands:

```
public static void createFile(FileSystem fs, String fileName)

public static void writeFile(FileSystem fs, String fileName, byte[] data)

public static void readFile(FileSystem fs, String fileName)

public static void removeFile(FileSystem fs, String fileName)
```

What happened when in touch commands

1. Call `public static void createFile(FileSystem fs, String fileName)`

To create a new file. .

2. Call `public static void writeFile(FileSystem fs, String fileName, byte[] data)`

To write to them.

```
fs.bitMap.nextClearBit(0);

fs.bitMap.set(freeBlock);

file.addBlock(freeBlock);

fs.blocks[freeBlock].writeData(Arrays.copyOfRange(data, dataOffset,
dataOffset + writeSize), 0);

file.setSize(file.getSize() + writeSize);
```

This two mode are call in this case of `handleCommand(fs, ... )` method

```
case "touch": {

    if (cmdPart.length == 2) {

        FileHandleHelper.createFile(fs, cmdPart[1]);

        break;

    } else if (cmdPart.length == 3) {

        FileHandleHelper.writeFile(fs, cmdPart[1], cmdPart[2].getBytes());

        break;

    }

    notValidCmd = true;

    break;

}
```